

Prediction of Chronic Disease Risk Using Nutrition, Lifestyle & Clinical Data from The Irish Longitudinal Study on Ageing (TILDA)



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Abstract

Background: Chronic diseases such as Cardiovascular Disease (CVD), Diabetes Mellitus (DM), Frailty, Gait, Exercise, Falls and Fractures (FR), and obesity represent major contributors to morbidity in older adults. While traditional prediction models focus heavily on clinical and functional indicators, the contribution of nutritional and lifestyle factors to early risk detection remains underexplored.

Methods: This thesis applies machine learning techniques to data from The Irish Longitudinal Study on Ageing (TILDA), a national representative longitudinal study of older adults. Five waves of socio-demographic, clinical, nutritional, psychological, and behavioural data, with selected variables from the externally produced Research and Development Corporation (RAND) + Harmonised TILDA dataset, together were systematically cleaned and standardised using a multi-stage preprocessing pipeline. Predictive models, including (*models x , y , z*) were developed to estimate the risk of chronic disease and FR-related outcomes. Model performance was evaluated using (*mention examples*), and feature importance was used to assess predictive value.

Results: TBD

Conclusions: TBD

Keywords: ageing, (Artificial Intelligence (AI)), Machine Learning (ML), predictive modelling, FR, chronic disease, nutrition, TILDA

Declaration

I Abderahman Haouit declare that I have produced this thesis without the prohibited assistance of third parties and without making use of aids other than those specified; notions taken over directly or indirectly from other sources have been identified as such. This thesis has not previously been presented in identical or similar form to any other Irish or foreign examination board.

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Your dedication goes here. Open the dedication.tex file found at:
0_frontmatter/dedication.tex

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GLOSSARY

Chapter 1

Introduction

Ageing populations experience substantial changes across physical, nutritional, cognitive, and psychosocial domains, each shaping vulnerability to chronic disease and functional decline [5–7]. Understanding how these multidomain factors interact over time is essential for early risk identification.

The financial and societal burden is substantial. In Europe, poor diet is a major risk factor for non communicable diseases, contributing billions in healthcare costs annually [8]. In Ireland, malnutrition alone is estimated to cost over €1.4 billion per year [9]. In response to these challenges, international public health frameworks, including the World Health Organization (WHO) *Global Action Plan* on noncommunicable diseases and Ireland’s *Healthy Ireland* framework highlights nutrition and lifestyle factors as central components of disease prevention strategies.

European regulatory frameworks such as the General Data Protection Regulation (GDPR) and proposed European Union (EU) Artificial Intelligence (AI) Act emphasise transparency and highlighting the need for interpretable Machine Learning (ML) models in healthcare.

Traditional risk prediction models have largely relied on clinical biomarkers such as Blood Pressure from Sphygmomanometer (BP) and lipid profiles. Yet, growing evidence shows that broader predictors including nutrition [10], Physical Activity (PA) [11], Scales and Questions on Mental Health and Wellbeing (MH) [12], and Social Variables (SOC) [13] improve predictive performance. Es-

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timating the likelihood of developing a condition given certain risk factors helps quantify the influence of these predictors in clinical decision making.

The Irish Longitudinal Study on Ageing (TILDA)¹ provides a national representative cohort of adults aged 50+ in Ireland, surveyed repeatedly across multiple domains. For cross study comparability and methodological consistency in model development, selected variables from the externally produced Research and Development Corporation (RAND) Harmonised TILDA dataset (available for Waves 1 and 2) were included.

The harmonised dataset provides standardised variable definitions and documented construction rules, which are particularly relevant in a machine learning setting where models are trained and evaluated on partitioned data. Its inclusion supports robust feature engineering, reproducible train validation splits, and alignment with prior ageing related ML studies that adopt harmonised frameworks. The specific role of harmonised variables within the modelling pipeline is detailed in Chapter 3.

Despite these advantages, existing ML research on chronic disease prediction faces persistent limitations: over reliance on clinical predictors, limited incorporation of lifestyle and SOC variables, predominantly cross sectional study designs that fail to capture trajectories, and a lack of interpretability consistent with emerging AI governance standards.

This thesis addresses these gaps by leveraging longitudinal TILDA’s data across multiple domains, using interpretable ML methods to combine modifiable health behaviours, psychosocial factors, and established clinical measures to enhance prediction of ageing related outcomes.

1.1 Research problem and motivation

Despite increasing research and clinical interest in lifestyle and nutrition factors, and advances in ML methods, approaches to chronic disease prediction in older adults remain limited in several ways.

¹TILDA’s dataset are pseudonymised and accessed under license via Irish Social Science Data Archive (ISSDA). Use adheres to UL S&E ethics approval and data retention requirements.

- **Limited Integration of Multidomain Predictors:** Existing models focus primarily on clinical measurements and often neglect modifiable behavioural, dietary, and psychosocial factors [10, 14].
- **Underuse of Longitudinal Dynamics:** Few studies leverage repeated measures to capture developments of risk, such as Frailty, Gait, Exercise, Falls and Fractures (FR) progression [15]. Probabilistic reasoning can help estimate the likelihood of developing a condition at a future time given earlier measurements, for example calculating the probability $P(\text{FR at wave 5} \mid \text{UGS at wave 1, Nutrition at wave 2})$.

This approach helps quantify how early indicators influence later outcomes and supports better feature selection and timing of interventions in predictive models.

- **Lack of Explainability:** Many ML models are not interpretable, limiting clinical adoption and failing to meet emerging regulatory requirements for trustworthy AI in healthcare [16].

Addressing these challenges is essential to improve predictive accuracy, support evidence based interventions, and ensure alignment with evolving policy and regulatory priorities.

1.2 Research aims and objectives

The goal of this thesis is to develop and validate ML models that integrate nutrition, lifestyle, and MH features with clinical data to improve prediction of chronic disease and ageing related outcomes in a representative cohort of older adults. This goal will be achieved through the following objectives:

- Review existing applications of ML for chronic disease prediction in ageing populations, with emphasis on gaps in lifestyle and nutrition integration.
- Develop a reproducible pipeline to preprocess and align TILDA waves 1–5, and engineer multidomain features (nutrition, activity, mobility, MH, clinical indicators).

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- Develop and compare baseline statistical models (e.g., logistic regression) with advanced interpretable ML methods such as Random Forest (RF) and Gradient Boosting Machine (GBM).
- Evaluate models using predictive classification performance metrics including Area Under the ROC Curve (AUC), calibration, and error measures (accuracy, precision, Recall (true positive rate) (Recall), F1-score (harmonic mean of precision and recall) (F1)), perform ablation analyses to assess the incremental value of lifestyle and nutrition predictors.
- Identify actionable predictors (e.g., vitamin D status, UGS reserve) and highlight modifiable risk factors for intervention.
- Document ethical, governance, and reproducibility practices for handling longitudinal pseudonymised health data.

1.3 Contributions of the research

This research makes contributions at three levels:

- **Empirical:** Demonstrates how behavioural, dietary, and MH features complement established clinical measures to enhance predictive performance.
- **Methodological:** Provides a reproducible pipeline for aligning longitudinal TILDA data across domains and integrating diverse features into ML workflows.
- **Clinical:** Highlights actionable modifiable factors alongside traditional clinical indicators for targeted interventions and policy.

1.4 Thesis Outline

In *Chapter 2* a literature review on ML prediction of chronic disease in ageing populations, with emphasis on lifestyle, nutrition predictors and methodological considerations is presented. *Chapter 3* discusses the TILDA study design, data

preprocessing, alignment, and feature engineering. In *Chapter 4* the modelling framework is outlined. This includes baseline and advanced ML, hyperparameter tuning, validation, and evaluation metrics. *Chapter 5* shows the model result, including model comparisons, assessments of feature importance, and an ablation analyses. In *Chapter 6* the implications, limitations, ethical considerations are discussed, and future directions for the research proposed. Additional supporting documents are included in appendices A.

Chapter 2

Literature Review

This chapter reviews literature on ageing, chronic disease, and predictive modelling using TILDA (introduced in Chapter 1). It focuses on three interrelated domains central to later life health: FR, physical function, MH, and nutrition within the context of biological ageing. This review focuses on FR, MH, and nutrition as key determinants of later life outcomes. In line with the WHO healthy ageing framework, research using TILDA highlights FR, MH, and nutrition as interconnected determinants of later life outcomes. The following subsections examine each domain in turn.

2.1 Methodology

Several sources describe how systematic and integrative reviews are commonly carried out in fields such as software engineering, health, and education. The approach followed key principles for systematic mapping, for example defining research questions, identifying relevant studies through structured database searches, applying inclusion and exclusion criteria, and categorising results as outlined in [1]. The integrative literature review [17] helped structure and connect findings across studies, while [18, 19] guided the organisation of ideas. In related methodological work, [2] clarified how Systematic Review (SR)s can identify research gaps, and [20] showed how mapping and synthesis highlight trends and future directions. This combined guidance informed a structured yet flexible methodology suitable for ageing in AI, and health research.

2.1.1 Search strategy and database screening

The main topic of interest was chronic disease prediction using nutrition, lifestyle, and clinical data from TILDA. Relevant databases (e.g., ScienceDirect, PubMed) were searched using combinations of keywords such as “Irish Longitudinal Study on Ageing”, “machine learning,” “chronic disease,” “nutrition,” “diet,” and “physical activity” (see table 2.1).

The search covered studies published from 2011 to 2025. Figure 2.1 summarises the number of studies identified per year, highlighting trends and gaps over time.

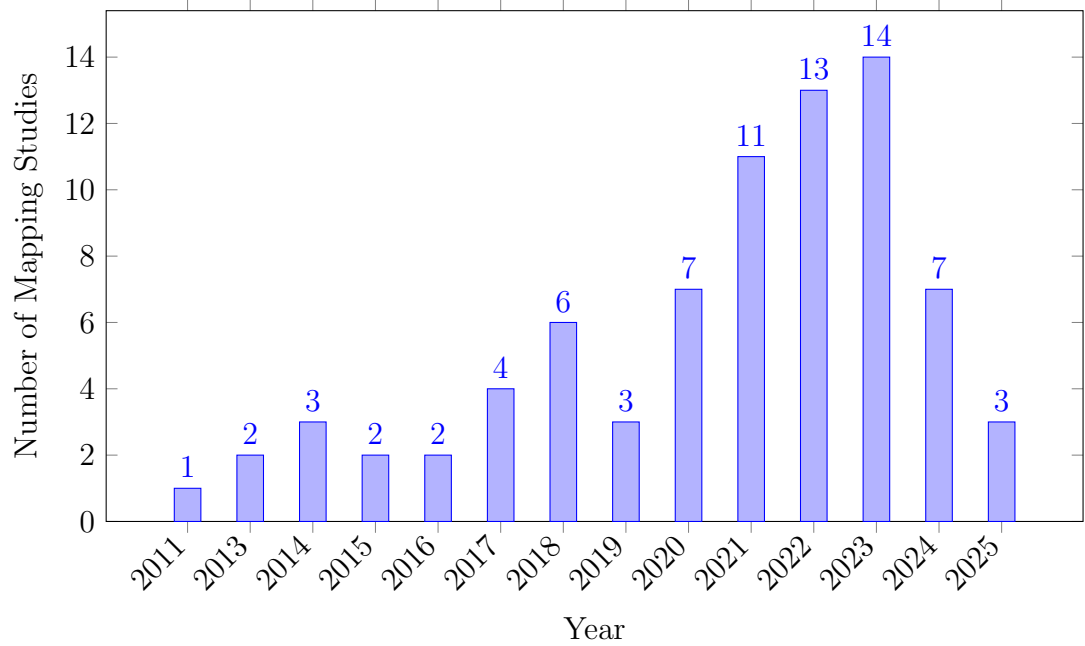


Figure 2.1: Publications per year.

Some search strings returned few or irrelevant results, particularly for ML related TILDA studies, highlighting a key research gap. Iterations continued until suitable combinations consistently returned relevant studies. Duplicates were removed using Zotero’s reference management feature. The remaining studies were individually and manually checked for relevance to “ageing,” “chronic disease,” and ML. Using Zotero’s notes functionality, the initially created table 2.3 captured key elements for each paper, including model type, features used, and

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Table 2.1: Database searches and number of studies per database 2.3 [1].

Database	Search	Results
Oxford Academic	(TILDA OR Irish Longitudinal Study on Ageing)AND (chronic disease OR diabetes OR cardiovascular) AND (nu- trition OR diet OR “physical activity”)	1081
	(TILDA OR “Irish Longitudinal Study on Ageing”) AND (“machine learning” OR “predictive model” OR “risk pre- diction”)	71
IEEE Xplore	TILDA OR “Irish Longitudinal Study on Ageing”	56
	(TILDA OR ”Irish Longitudinal Study on Ageing”) AND (“machine learning” OR “predictive model” OR “risk pre- diction”)	7
ScienceDirect	(TILDA OR “Irish Longitudinal Study on Ageing”) AND (“chronic disease” OR cardiovascular OR diabetes) AND (nutrition OR diet OR lifestyle OR “physical activity”)	257
	(TILDA OR “Irish Longitudinal Study on Ageing”) AND (“machine learning” OR “predictive model”) AND (“chronic disease” OR diabetes) AND (nutrition OR “physical activ- ity”)	20
PubMed	(TILDA OR “Irish Longitudinal Study on Ageing”) AND (“chronic disease” OR cardiovascular OR diabetes) AND (nutrition OR diet OR lifestyle OR “physical activity”)	27
	(TILDA OR “Irish Longitudinal Study on Ageing”) AND (“machine learning” OR “predictive model” OR “risk pre- diction”)	9

main findings for each paper in the collection, as illustrated in the PRISMA flow diagram (Figure 2.2).

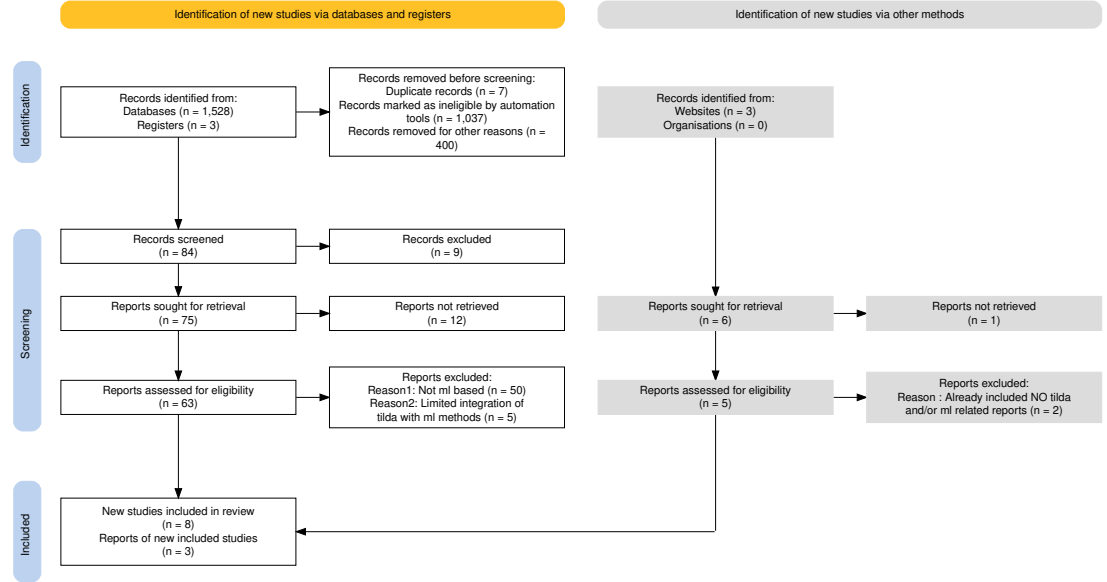


Figure 2.2: PRISMA chart for SRs showing the study selection process for the systematic mapping and integrative review [4].

2.1.2 Quality and relevance evaluation

To systematically assess the quality of the included studies, a customisable version of Newcastle-Ottawa Scale (NOS) was applied, see table 2.2. The NOS evaluates studies across three domains: *Selection* (S1–S4), *Comparability* (C1–C2), and *Outcome* (O1–O3). *Selection* criteria assess the representativeness of the sample, clarity of exposure measurement, and temporal alignment of outcomes. *Comparability* examines the control of key confounders (variables that may influence both the predictor and outcome), while *outcome* evaluates measurement validity, follow up adequacy, and completeness. Each domain is scored 0–1, recall the Bernoulli random variable (0 = criterion not met, 1 = criterion met), and the total score reflects the overall study quality.

This initial table guided a further refinement of the results into two sub tables: (1) studies critically related to TILDA, (2) studies using TILDA data with ML

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for predictive contexts, which revealed minimal to no integration with TILDA.

Table 2.2: Quality evaluation of Systematic Literature Review (SLR)s [2]

Reference	S1	S2	S3	S4	C1	C2	O1	O2	O3	Total
(Xu, 2023 [21])	1	1	1	1	1	1	1	1	1	9
(Donoghue, 2023 [16])	1	1	1	1	1	1	1	0	1	8
(Moguilner, 2021 [15])	1	1	1	1	1	0	1	1	1	8
(Davis, 2021 [5])	1	1	1	1	1	0	1	1	0	7
(Boyle, 2021 [22])	1	1	1	1	1	0	1	0	1	7
(Tzemah-Shahar, 2022 [23])	1	1	1	0	1	0	1	1	1	7
(A. Shirsath, 2024 [24])	1	1	1	1	1	1	1	0	1	8
(Romero-Ortuno, 2021 [25])	1	1	1	1	1	1	1	1	1	9
(Bardon, 2020 [26])	1	1	1	0	1	0	1	1	1	7
(Zhang, 2025 [27])	1	1	1	1	1	1	1	0	1	8
(Sutin, 2023 [28])	1	1	1	1	0	0	1	1	1	7

Table 2.3 summarises selected studies on ageing using TILDA and ML, reflecting key gaps and opportunities identified during systematic mapping. These steps were aligned with the research matrix created at the beginning, ensuring that the refinement of studies and synthesis of findings stayed closely tied to the original research goals.

2.1.3 Synthesis and Integration

Using the NOS scores, each study was annotated in Zotero with details on dataset, key variables, analytical approach, and main findings, facilitating a consistent and systematic evaluation across studies. For the systematic component, data was systematically extracted and synthesised to identify trends (e.g., Markov models, logistic regression, 5-fold Cross Validation (CV)) and research gaps [2], complemented by DoctoralWriting resources [29, 30], which clarified how to define research questions and develop logical arguments.

The integration of a research matrix with systematic mapping and integrative synthesis ensured that the methodology was well suited to capturing the complexity of ageing, nutrition, chronic disease risk and predictive modelling studies focused primarily on TILDA based studies, but due to the limited number of ML

2.1 Methodology

Table 2.3: Summary of selected studies on ageing using TILDA

Theme	Author/year	Dataset	Variables/features	Methods/model type	Key Findings/outcome	Limitations
FR	Romero-Ortuno (2021) [25]	Wave 1–5	FR state, FR phenotype (FP) components (Exhaustion(EX), Unexplained weight loss(WL), Weakness(WK), Slowness(SL), Low physical activity(LA))	Multi state Markov model, longitudinal 8 year follow up	32% of pre frail to non frail; FR strongly predicted 31% mortality	Self report bias; observational design
Falls + ML	Donoghue (2023) [16]	Wave 1–3	Gait, mobility, medications, history of falls	Conditional inference forest; 5-fold CV; undersampling for imbalance	Accuracy up to 69.7%; for adults aged 50–64.	Area Under the Receiver Operating Characteristic Curve (AUROC) values were ≤ 0.69 and PRAUC ≤ 0.39 indicating low predictive accuracy; no external validation; cross sectional; Class imbalance
Fear of Falling (FOF)	Zarudsky (2014) [31]	Wave 1	Age, sex, fall history	Cross sectional prevalence analysis	23.3% prevalence; higher in women; 70% with FOF had no recent falls in the last 12 months	Self reported; cross sectional
Functional Mobility	Huang (2019) [32]	Wave 1–2	Measured by Timed Up and Go (mobility test) (TUG), Hand Grip Strength (HGS), physical activity, vision, Body Mass Index (BMI), age	Linear regression, stratified analyses	Age modifies effects; PA and grip protect mobility; vision benefit only in middle aged (Beta = -0.032, P = 0.002)	Observational; 2 year follow up
Depression	Jacob (2023) [33]	Wave 1–2	Falls, FOF, HADS-A, Center for Epidemiologic Studies Depression Scale (CES-D)	Prospective logistic regression	Falls predicted incident anxiety Odds Ratio (OR)=1.58 and depression OR=1.43	Self reported falls; short follow up; mediation effect partially explored
	Laird (2023) [14]	Wave 1–5	Moderate-to-Vigorous Physical Activity (MVPA), walking, sociodemographics	Longitudinal regression; mixed model	Higher physical activity reduced depressive symptoms; dose response effect	Self reported PA; observational; limited causal inference
Nutrition	Murphy (2023) [10]	Wave 1–5	Plasma lutein, zeaxanthin	Linear/ordinal logistic regression	Higher carotenoids pre-linked to slower FR progression Lower odds of becoming frail (ORs = 0.57–0.89)	Observational; not predictive of longitudinal muscle changes
	Bardon (2020) [26]	Wave 1–2	Hospitalisation, functional status, SOC support, sex, falls	Logistic regression; sex stratified analyses	Functional decline, hospitalisation, poor support increased malnutrition risk; sex differences	Observational(short follow up (2 years)); limited generalisability ;65 years, self reported
ML	Xu (2023) [21]	Wave 1–3	105 predictors from 40,000+ variables (SOC, clinical, mental, family)	Stacked ensemble: RF, XRT, GLM, GBM, Deep Neural Network (DNN); Shapley Additive Explanations (SHAP) explainability	AUC=0.854 for 2 year Diabetes Mellitus (DM) prediction; top predictors LDL, cirrhosis, sex	No external validation; short follow up; ensemble complexity limits deployment
	Davis (2021) [5]	Wave 3	Age, HGS, chair stands, BMI, cognitive tests	Histogram gradient boosting regression; SHAP	r^2 test: UGS $\approx$ 0.41, Maximum Gait Speed (MGS) $\approx$ 0.46, GSR $\approx$ 0.21	Modest predictive power; cross sectional; no external validation
	Zhang (2025) [27]	Longitudinal IMAGEN cohort	Psychopathology, personality, cognition, substance use, environment	Regularised logistic regression; diagnostic	Longitudinal AUC: ED=0.71, MDD=0.64, AUD=0.67; shared transdiagnostic predictors	Limited age range; no external replication
	Boyle (2021) [22]	Wave 3	Brain predicted age difference (brainPAD); processing speed; attention; cognitive flexibility; verbal fluency	Linear regression, correlation analyses	Higher brainPAD associated with lower cognitive performance across multiple domains	Cross sectional; no causal inference; limited sample

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applications within TILDA, additional studies on older adult populations and chronic disease were included.

2.2 Frailty and physical function

FR reflects declining physiological reserve, providing a framework for understanding physical vulnerability in ageing. In TILDA, it is captured through both the FR phenotype and the FR index, with UGS, HGS, and mobility decline serving as practical predictors. These measures strongly predict hospitalisation and mortality.

2.2.1 Probabilistic interpretation of frailty

One can argue that from a probabilistic perspective, the chance of an adverse outcome (such as mortality, Y) given observed features (such as FR markers, X) as a conditional probability $P(Y | X)$. This allows us to deduce which FR features are most predictive of adverse outcomes. Features may be independent, dependent, or mutually exclusive.

The presence or absence of metabolic syndrome represents a mutually exclusive event that can inform risk models.

$$P(X_i \cap X_j) = P(X_i)P(X_j)$$

for independent features X_i and X_j , guiding feature selection and reducing redundancy in model inputs. This reflects the likelihood that an individual with certain FR characteristics will experience a particular outcome.

Evidence suggests that FR is dynamic, not irreversible. For example, multi state modelling showed that many prefrail adults transition back to non frail, although established FR strongly predicts mortality [25]. Metabolic syndrome accelerates the development of FR, suggesting a metabolic pathway of vulnerability [34]. The presence of metabolic syndrome modifies the conditional probability $P(\text{FR} | \text{Metabolic Syndrome})$ and may increase the likelihood of adverse outcomes over time. Another detail is that age matters: obesity predicts mobility decline earlier in life, whereas later in life, traits that protect against this, including HGS, become more significant [32].

These findings partly conflict: FR appears both reversible and progressive, depending on risk context. This complexity highlights the need for probabilistic models that can update beliefs about risk as new features are observed, as formalised by Bayes theorem:

$$P(Y | X) = \frac{P(X | Y)P(Y)}{P(X)}$$

where $P(Y | X)$ is the posterior probability of outcome Y given features X .

2.2.2 Bayesian updating of frailty risk

Causal inference is limited by observational designs, and predictors are frequently examined separately, ignoring interactions. Most FR models have only been with the TILDA dataset and with this study that focuses on the same TILDA but uses both longitudinal TILDA waves and harmonised variables to improve cross study comparability, enabling for a more robust training validation splits in ML models, and explores richer predictive patterns over time. These gaps point to the need for multidimensional approaches. Traditional regression models may not fully capture age dependent interactions, suggesting a role for more flexible methods such as machine learning techniques and models.

2.3 Mental health in older adults

MH, including depression, anxiety, loneliness, and SOC isolation, strongly shapes quality of life. Evidence from TILDA shows strong links to both physical and SOC factors. [35] found that participation in physical, SOC, and cognitive activities summarised by the Act-Belong-Commit (ABC) indicator was associated with lower depression and anxiety. This suggests SOC integration and lifestyle protect mental wellbeing. However, [36] reported that self reported measures may exaggerate such associations..., raising concerns of bias. These two findings conflict: the effect of SOC engagement is strong when self reported, weaker when objectively assessed.

Objective data strengthen the case for activity. Accelerometer based measures show a dose response link between moderate to vigorous activity and lower

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depressive symptoms [11]. Yet reverse pathways are also evident: incident falls and functional decline predict later depression, refereed by FOF [33]. Together, these results reveal a bidirectional relationship, MH both influences and is influenced by physical decline. Most studies rely on broad self report scales such as CES-D or Hospital Anxiety and Depression Scale (HADS), with little biomarker or clinical validation.

This literature suggests MH in ageing cannot be modelled as a one way effect. It requires longitudinal, multidimensional approaches that can capture mutual pathways, an area where machine learning is well suited especially when combined with probabilistic models that estimate the likelihood and posterior probability of MH outcomes given multiple interacting features.

For example, we can estimate:

$$P(\text{Depression} \mid \text{Activity, Falls, SOC Engagement}),$$

quantifying how these interacting features together influence MH outcomes. Probabilistic modelling helps determine which features have the strongest effect and which may be redundant for predictive purposes.

2.4 Nutrition and biological ageing

Nutrition interacts with both biological and psychosocial systems, shaping ageing trajectories and FR. Biomarkers like vitamin D, lutein, and zeaxanthin have been linked to physical and cognitive outcomes in TILDA. [10] showed that higher plasma lutein and zeaxanthin dietary carotenoids found in leafy vegetables were linked to slower FR progression. By contrast, [37] found vitamin D deficiency predicted DM, musculoskeletal decline, especially among adults with low sunlight exposure. These studies point to different nutritional mechanisms: antioxidant protection versus metabolic regulation.

2.4.1 Bayesian perspective on nutritional biomarkers

The effect of nutrition on ageing outcomes can be formalised probabilistically using Bayes theorem:

$$P(\text{Outcome} \mid \text{Nutritional Status}) = \frac{P(\text{Nutritional Status} \mid \text{Outcome})P(\text{Outcome})}{P(\text{Nutritional Status})},$$

where the likelihood function $P(\text{Nutritional Status} \mid \text{Outcome})$ quantifies how observed nutritional biomarkers relate to health states, and the posterior $P(\text{Outcome} \mid \text{Nutritional Status})$ captures updated beliefs about outcomes, including uncertainty in small sample studies. This helps identify which biomarkers are most informative for predicting ageing outcomes which supports feature selection in ML models.

2.4.2 Contextual and integrative considerations

Malnutrition also reflects SOC and functional risks. [26] found that hospitalisation, functional loss, and poor SOC support increased malnutrition risk, with notable sex differences. This highlights a limitation in much nutritional research: biological markers are studied without integrating SOC context.

The difference between biological and chronological age is a major point of debate. Biomarker based metrics that better capture these cumulative exposures, like FR indices or epigenetic clocks, frequently vary from chronological age. McCarthy and Azizi link metabolic syndrome and nutritional deficits to accelerated biological ageing [38, 39]. Although the majority of the evidence is cross sectional and has little ability to capture changes over time, these indicators offer a more precise prediction of vulnerability. Reliability is also decreased in biomarker research with small sample sizes, which can be formalised through the likelihood function $P(\text{Data} \mid \theta)$ and its impact on posterior uncertainty in Bayesian inference.

Taken together, nutrition interacts with biological, metabolic, and SOC systems. Isolating single nutrients risks oversimplification. Integrative modelling that combines biomarkers, clinical factors, and SOC determinants supported by machine learning pipelines offers a more promising route. Models that estimate

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the joint distribution of features and outcomes can better capture the complex dependencies relevant to biological ageing.

2.5 Machine Learning in ageing and chronic disease

ML is increasingly applied in ageing research because it can capture non linear and multidimensional relationships that traditional regression models, such as logistic regression, often miss [21, 22].

2.5.1 Feature sets in ML studies

Studies using ML in ageing research include many types of predictors: clinical measures, sociodemographic factors, lifestyle habits, cognitive tests, and psychosocial variables. A recurring pattern, however, is that models often underline easily measured physical health data, such as comorbidity counts, mobility scores, or vital signs, while more complex but important predictors, including nutrition and SOC factors, are underrepresented [31]. This imbalance may limit the ability of models to capture the full complexity of chronic disease risk with ageing, especially considering that MH and nutrition exert independent and interacting effects on later life outcomes (here). Models that focus only on ready available clinical data risk reproducing the same biases seen in traditional regression approaches.

2.5.2 ML models used

The methodological approaches used in ageing research span both traditional and advanced models. Logistic regression and Markov models remain common baselines due to their interpretability and simplicity, yet they are often unable to capture non linearities or higher order interactions. More recent work has shifted towards ensemble methods such as RF, gradient boosting machines GBM, as well as DNNs, which generally achieve higher predictive accuracy.

Similarly, [40] evaluated mortality prediction using a standard logistic regression model with demographic, clinical, and functional predictors, achieving an

AUC of 0.78, whereas, ensemble methods in [21] reached 0.854 for DM prediction, showing how flexible, non linear models can better capture complex interactions among ageing related predictors.

2.5.3 Performance and Validation

Predictive performance across studies is generally moderate to strong, with AUC values between 0.70 and 0.90 for outcomes like DM, UGS, depression, and mortality. This indicates that models can distinguish reasonably well between individuals who will or will not experience an outcome.

However, there are important limitations. Many studies rely only on internal validation, testing models on the same data used for training. This can inflate performance estimates and reduce confidence in applying the model to new populations. To add, most studies are cross sectional, so causal or temporal effects cannot be assessed. When combined with the underuse of nutrition and SOC predictors, these issues indicate that current models may not fully represent the multidimensional nature of ageing and could produce context specific results rather than generalisable predictions [31].

2.5.4 Explainability in ML for ageing

As ML models become more complex, interpretability emerges as a critical challenge. Explainability tools like SHAP can highlight feature importance, but their use is inconsistent, particularly in deep learning models [5, 21, 41]. These methods improve transparency and can highlight unexpected drivers of outcomes, such as psychosocial variables being stronger predictors than biomedical ones, yet black box concerns remain, especially for deep learning models, where feature interactions are difficult to track. For ageing research, where clinical adoption requires both accuracy and trust, explainability is not optional but essential. Current efforts show promise but remain fragmented, with no consensus on best practice. This gap links directly to the broader methodological limits discussed in the next section on gaps in the literature.

2. LITERATURE REVIEW

2.6 Gaps in the Literature

First, most studies rely on TILDA or similar cohorts of Irish and predominantly older adults (50+). This limits confidence in how well results transfer to other populations. This thesis focuses on TILDA to study ageing in depth but also makes use of the harmonised dataset, which links TILDA with comparable UK and US cohorts. This helps place findings in a broader international context while retaining a core emphasis on older adults.

Second, there is an over reliance on physical health predictors such as comorbidity counts and mobility measures, while nutrition, SOC, religious or spiritual, and environmental variables are often overlooked. As shown in [10], nutritional biomarkers like lutein and zeaxanthin are strong predictors of FR, yet these are rarely integrated into broader ML pipelines. By contrast, studies such as [26] highlight the role of SOC support and functional decline in malnutrition risk. Together, these examples show that excluding nutrition and SOC factors risks producing models that reinforce the same biomedical biases seen in regression heavy work. From a Bayesian viewpoint, the omission of key features reduces the explanatory power of the model and weakens the posterior probability $P(\text{Outcome} \mid \text{All Features})$ compared to $P(\text{Outcome} \mid \text{Limited Features})$. Including a richer set of features increases the likelihood that the model accurately reflects the true data generating process.

Third, methodological choices remain conservative. Regression models remain dominant, likely due to their interpretability and ease of implementation. However, they often fail to capture non linear interactions and higher order dependencies present in ageing related data. Comparisons with more advanced approaches such as DNNs or transformers are rarely explored. For instance, [40] achieved an AUC of 0.78 using logistic regression for mortality prediction in TILDA, whereas [21] reported an AUC of 0.854 with ensemble methods for DM incidence. This contrast illustrates that more flexible ML models can provide superior predictive performance while maintaining interpretability when combined with explainability techniques. Where else ensembles or brainPAD approaches have been attempted, validation is almost always internal, with little cross cohort testing. Weak validation and limited feature sets reduce the generalisability of the

estimated conditional probabilities and increase the risk of overfitting. Maximum likelihood estimation and Bayesian inference can provide a formal framework for assessing model fit and updating predictions as more diverse and representative data become available.

Fourth, many outcomes relevant to older adults such as pain, sleep quality, and subjective wellbeing remain underexplored. For example, [10] focused on plasma biomarkers for FR without assessing sleep or pain, and UGS models such as [5] similarly omit subjective quality of life. Omitting these outcomes limits the real world application of predictive models. Including broader outcomes in ML pipelines can better capture holistic ageing trajectories and improve interpretability for interventions targeting older adults.

Fifth, cross sectional bias and data handling weaken reliability. Much nutritional work, such as [10], relies on single time point biomarker data, which cannot capture longitudinal change. By contrast, studies applying multi wave TILDA data [25] demonstrate the added value of modelling temporal dynamics. Similarly, missing data are often handled through list wise deletion, as in [40], which risks biasing results toward healthier participants, whereas more robust imputation approaches remain underused. Missing or poorly handled data can distort the likelihood function and hence the posterior predictions, reducing confidence in the model’s outputs.

Finally, interpretability challenges persist. While tools like SHAP have been applied to highlight feature importance [21, 41], their use is inconsistent, and consensus on best practice has yet to emerge. Without transparent explanations, even accurate models may face barriers to clinical acceptance. Taken together, these examples show that ageing research continues to rely on narrow predictors, conservative models, and weak validation. Addressing these gaps requires integrative, diverse, and transparent approaches that link multidimensional predictors with explainable ML.

2.7 Chapter Summary

This chapter reviewed evidence on ageing and chronic disease across three domains, FR, physical function, MH, and nutrition and biological ageing. We also

2. LITERATURE REVIEW

examined how machine learning has been applied to predict later life outcomes. The findings showed that while FR is partly reversible, it remains strongly tied to mortality, that MH both shapes and is shaped by physical decline, and that nutrition interacts with biological, metabolic, and SOC systems in ways that are often underestimated.

While machine learning has improved prediction accuracy in ageing research, challenges remain in integrating diverse predictors and ensuring interpretability and longitudinal validation.

Addressing these challenges requires the integrative, longitudinal, and transparent approaches that combine diverse predictors. These gaps motivated the feature engineering and modelling strategies implemented in Chapters 3 and 4, including the use of harmonised variables to ensure reproducibility, generalisability, for ML modelling.

Chapter 3

Technical Approach

3.1 Study Design

TILDA provides longitudinal data across five waves for community dwelling adults in Ireland. This thesis uses data from five core TILDA waves (Waves 1–5). For contextual comparison and benchmarking against international cohorts (see Chapter 1) however, it is important to note this is an external resource and not the output of the thesis’s primary preprocessing pipeline. The primary analytical work focuses on the full five wave TILDA dataset.

3.2 Cohort Definition

All participants who provided data in at least one TILDA wave were considered eligible. To maximise the available sample for machine learning, no exclusions were applied on the basis of age, medical conditions, mobility, cognitive status, or health centre attendance.

Participants were removed only when, after the combined manual filtering and automated preprocessing pipeline, they had no remaining usable variables. Filtering therefore occurred primarily at the *variable level*, with features removed for extreme missingness, lack of variation, or non informative structure (see Chapter 4). This approach preserves the largest possible sample while ensuring all retained inputs are analytically meaningful.

3. TECHNICAL APPROACH

3.2.1 Cohort inclusion criteria

- **Participant inclusion:** All respondents who contributed data in at least one wave were included in the initial sample.
- **Usable data requirement:** Participants were excluded only when, after manual filtering and preprocessing, no usable variables remained.
- **Longitudinal retention:** Individuals were retained regardless of how many waves they contributed to, enabling initial maximum dataset utilisation.
- **No clinical exclusions:** No participant was removed due to medical diagnoses, cognitive scores, functional limitations, or non attendance at the health centre assessment.

This approach prioritises data integrity and sample size for machine learning applications rather than clinical eligibility criteria common in classical epidemiological analyses.

3.3 Data Collection

Data used in this thesis were collected across five waves:

- **Wave 1:** 2009–2011
- **Wave 2:** 2012–2013
- **Wave 3:** 2014–2015
- **Wave 4:** 2016
- **Wave 5:** 2018–2019

All TILDA data were pseudonymised prior to release, meaning that personal identifiers such as names and addresses were removed or replaced with unique codes to protect participant confidentiality. Each participant received a unique

identifier ('id'), along with household and cluster identifiers ('household', 'cluster'), allowing longitudinal linkage across waves without revealing individual identities.

Pseudonymisation ensures that analyses comply with data protection standards while retaining the structural and analytical integrity of the dataset. From a data cleaning and machine learning perspective, pseudonymised identifiers enable accurate merging of data across waves, preserve the longitudinal design, and support reproducible preprocessing and feature engineering. While these identifiers cannot be used for analyses of personal demographics like names or addresses, this limitation does not affect the modelling of health outcomes, lifestyle factors, or other variables of interest.

Each wave included three primary components:

1. **Computer Assisted Personal Interview (CAPI):** Home based, computer assisted personal interviews covering health, socio demographics, and family.
2. **Self Completion Questionnaire (SCQ):** Paper based self completion questionnaires capturing sensitive information on mental health, lifestyle, and wellbeing.
3. **Health Assessment:** Clinical and biomarker measurements conducted in dedicated centers or at home (Wave 1 and Wave 3).

3.3.1 Data Overview

Before processing, raw TILDA datasets were manually inspected, and clear non informative variables (e.g., administrative identifiers, constant columns, high missingness) were removed. Table 3.1 shows the raw dimensions of each dataset.

3.3.2 Variable filtering criteria

Manual filtering used rule of thumb prior to automated cleaning:

3. TECHNICAL APPROACH

Table 3.1: Dimensions of Raw TILDA and Harmonised Datasets (Pre Filtering)

Dataset	Participants (N)	Raw Variables (N)
Wave 1	8501	1619
Wave 2	7206	724
Wave 3	6397	1029
Wave 4	5713	990
Wave 5	4978	983
Harmonised	8501	572

- **Missingness:** Drop variables with $> 80\%$ missing values. This threshold balances retaining sufficient feature coverage with ensuring statistical reliability. Variables exceeding 80% missingness provide limited information in most cases, increase imputation uncertainty, and reduce effective sample size, which is particularly critical in longitudinal machine learning analyses.
- **Unique values:** Drop variables with a single observed value, as they contain no discriminative information and cannot contribute to modelling or inference. Variables with low cardinality (2–4 unique values) were retained for review, as they frequently represent meaningful categorical responses (e.g., binary indicators or ordinal scales) relevant to ageing outcomes.
- **Dominant category:** Flag variables with extremely frequent categories (e.g., $> 90\%$) for review.
- **Identifiers / administrative fields:** IDs, version tags, and sampling weights were identified using the Pseudonymised Microdata File (PMF) metadata. Variable names and descriptions were exported from each wave’s PMF file into CSV files (code provided in Appendix A) to systematically document and review features prior to filtering. These exports enabled structured manual inspection and traceability between raw PMF metadata and downstream feature selection decisions.

These manual checks produced a reduced variable set, subsequently processed using the automated nine scenario cleaning pipeline (Chapter 4).

3.3.3 Variable filtering flowchart

As part of this workflow, PMF metadata exports were first reviewed in spreadsheet form, where variables deemed non informative were visually flagged (e.g., marked in red). Candidate variables retained after manual review (described in 3.3.2) were consolidated using spreadsheet formula (e.g., `=TEXTJOIN(", ", TRUE, J2:Jn)`), which concatenated all non excluded variable names into a comma separated list. Here, n is the final row index of retained variables, with J2 corresponding to the first data row below the column header. These lists were manually transferred into wave specific JavaScript Object Notation (JSON) files, which were subsequently used by the Python pipeline to selectively load and filter only the retained variables from the original Statistical Package for the Social Sciences (SPSS) (`.sav`) and Stata (`.dta`) data files. This procedure ensured a reproducible and traceable workflow from manual selection to automated preprocessing, as illustrated in the flowchart (Figure 3.1).

3.4 Thematic Grouping

After manual filtering, retained variables were organised into six thematic domains aligned with ageing literature and study objectives. The grouping process followed the standard PMF naming conventions (Table 3.2). Specifically, each variable's prefix indicated its data source or domain, which allowed to systematically assign variables to one of six thematic domains. This ensured that features included in each domain were conceptually coherent and analytically meaningful.

3.4.1 Derived variable prefixes

To systematically process TILDA variables, each variable was assigned a prefix indicating its data source or research domain. Variables collected through the home based CAPI follow a two letter module code plus three digit question code format (e.g., `ph121`). For questions with multiple responses or loops across participants or objects, an underscore and number indicates the specific response or loop iteration (e.g., `ph201_05`).

3. TECHNICAL APPROACH

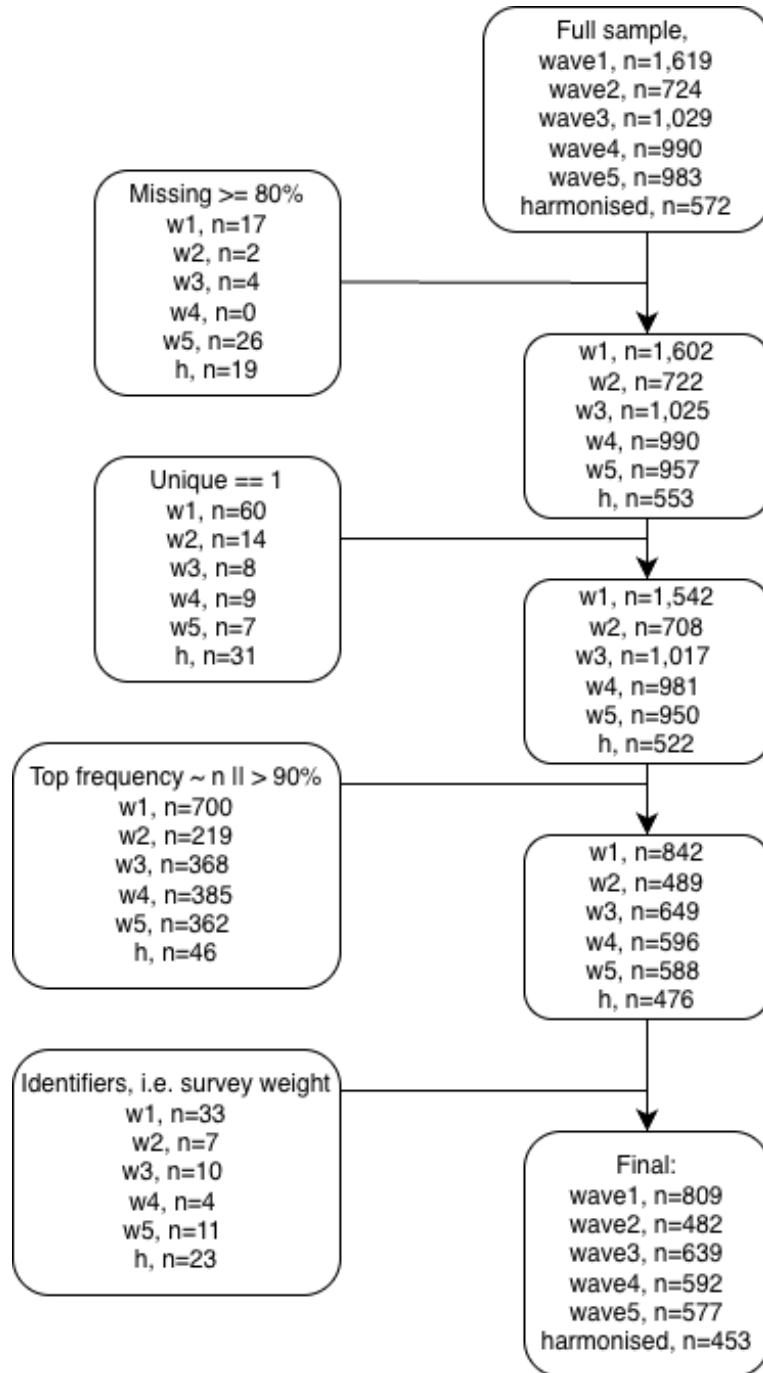


Figure 3.1: Flowchart of the manual variable filtering criteria. Placeholder counts indicate remaining variables per dataset after each step.

Variables from the self completion questionnaire (SCQ) are prefixed with ‘SCQ’. Derived variables, constructed from either CAPI or SCQ data, use an uppercase code corresponding to the research area (Table 3.2). These prefixes provide a consistent framework for categorising variables and guided the systematic assignment of features into thematic domains for downstream analysis.

Table 3.2: PMF Derived Variable Prefixes ([3])

Variable Prefix	Research Area
ADL	Activities of daily living
BEH	Health behaviours
BLOODS	Results of blood sample analysis
BP	Blood pressure from sphygmomanometer
COG	Scales and questions on cognition
CRT	Choice reaction time (cognitive test)
DIS	Disability, functional impairment, helpers
FR	Frailty, gait, exercise, falls and fractures
G2G	Gateway to Global Aging Data
GRIP	Grip strength
IADL	Instrumental activities of daily living
INC	Income variables
MD	Medication data
MH	Scales and questions on mental health and wellbeing
SCR	Screening tests
SES	Social class and socio
SOC	Social

3.4.2 Feature groups

Using the PMF prefixes as a guide, variables were systematically mapped to six thematic groups (Table 3.3). Each thematic group captures a conceptually coherent domain, reflecting physical, cognitive, social, and behavioural aspects of ageing. This categorisation served two key purposes in the thesis:

3. TECHNICAL APPROACH

1. It structured the data for the automated nine scenario cleaning engine (Chapter 4), ensuring consistent application of filtering rules and facilitating reproducible preprocessing.

2. It aligned feature selection with the analytic focus of the thesis, enabling machine learning models to leverage variables that are both conceptually meaningful and longitudinally robust across multiple waves.

The six thematic groups integrate information from CAPI, SCQ, and derived variables, covering socio demographic context, physical health, functional status, lifestyle behaviours, nutrition/metabolic biomarkers, and psychological/cognitive measures. This structured grouping enhances interpretability, ensures comprehensive domain coverage, and provides a foundation for downstream predictive modelling and risk assessment.

3.4 Thematic Grouping

Table 3.3: Thematic grouping of longitudinal features

ID	Thematic Group	Description	Representative Variables
1	Socio Demographic & Context	Baseline stratification factors	Age, Gender, Education, Income Variables (INC), Marital Status, SOC
2	Physical Health & Clinical Baseline	Self reported and clinical indicators	Self Rated Health Screening Tests (SCR), BMI, BP, Comorbidity Count Medication Data (MD)
3	Functional Status & FR	Measures of mobility and physical decline	FR Index FR, UGS, Falls History <i>SCQFalls</i> , Instrumental Activities of Daily Living (IADL) Disability, Functional Impairment and Helpers (DIS)
4	Lifestyle & Behavioural	Modifiable health related behaviours	Smoking, Alcohol Intake <i>SCQAlco</i> , PA International Physical Activity Questionnaire (IPAQ)
5	Nutrition & Metabolic Biomarkers	Objective nutritional and metabolic markers	Plasma Carotenoids, Lutein, Zeaxanthin, Vitamin D/B12, Cholesterol
6	Psychological & Cognitive	Mental health and cognitive performance	CES-D, Loneliness <i>MHucla</i> , Scales and Questions on Cognition (COG), Choice Reaction Time (Cognitive Test) (CRT)

Chapter 4

Implementation Details

4.1 Data Preprocessing

The preprocessing pipeline primary aim was to convert all features, regardless of their original datatype (float, categorical, object, or mixed formats), into a standardised numerical representation suitable for machine learning. All non response and sentinel values were consistently mapped to the numerical placeholder `-1.0`, and all valid entries were converted to `float64`. This ensured compatibility across the TILDA waves and enabled uniform downstream operations for feature engineering and modelling.

4.1.1 Input data shape (post manual filtering)

The dimensions of the manually filtered datasets, which serve as the input for the automated cleaning engine, are reported in (table 4.1). This initial manual filtering step removed non informative variables (e.g., columns with zero variation, unique participant IDs, non essential administrative fields) , resulting in a reduction of up to 50% of the original raw variables (e.g., from 1619 to 809 in Wave 1).

4.1.2 Cleaning Engine

The core challenge in converting heterogeneous variable types across five waves into a single numerical standard required constructing a reproducible pipeline to

Table 4.1: Input Shape for Cleaning Pipeline (Post Manual Variable Filtering)

Dataset	Row (participants)	Column (variables)
Wave 1	8501	809
Wave 2	7206	482
Wave 3	6397	639
Wave 4	5713	592
Wave 5	4978	577
Harmonised RAND	8501	453

achieve *numerical homogeneity*. This was accomplished by developing a two stage system: *Variable scenario mapping*, specifically inside “detect_column_scenario_mapping” function C and the final *six step deterministic cleaning engine* D.

Stage 1: Variable scenario mapping and validation

An exploratory visualiser was used to systematically inspect the raw data distributions, unique values, and datatype inconsistencies across all five waves. This audit initially identified between five and seven distinct data scenarios, which were expanded and refined during development to nine total scenarios to cover the full scope of data heterogeneity: from simple numeric fields and numeric data with symbols, to likert type scales, calendar based frequency strings, and complex mixed string numeric fields. These nine initial scenarios were then systematically collapsed and categorised into the six final cleaning steps, reflecting an ordering based on transformation confidence.

The pipeline’s first stage utilised an automated function (see appendix C), powered by the **re** (Regex) python library, to classify every raw variable into one of these nine scenarios. During development, a dedicated “unknown complex” category was crucial: variables routed here indicated a failure of the detection logic. This iterative debugging process, which identified and fixed classification errors, significantly drove the robustness of the final detection mechanism by ensuring the pipeline correctly classified variables that initially appeared ambiguous.

To ensure robustness, the classification output was immediately processed by a validation routine that generated a comprehensive dictionary of all variables,

4. IMPLEMENTATION DETAILS

specific to its respective grouped detected scenario after each step (used for manual inspection), their detected scenario, and a sample of transformed values. This dictionary generated inside the “show_grouped_vars_for_scenarios” function was formatted to create a automatic friendly “.md” file for rapid, surface level verification, which honed the final mapping dictionary D. This process of using automated detection followed by visual, manual validation was essential for correcting errors in classification logic before the subsequent scenario transformations were executed in the pipeline.

Stage 2: Deterministic six step cleaning sequence

The nine input scenarios were processed by the *Six step cleaning engine*. Resolving simple and unambiguous cases first increased the proportion of clean numerical variables before addressing complex structures, which involved the censored and multi-class scenarios (Figure 4.1). This deterministic sequence applied the same logic across all waves to ensure consistent data alignment (see appendix D).

A key difficulty in cleaning the five TILDA waves was the lack of guaranteed ordering in categorical values for the same variable across different waves (e.g., Wave 1: [Yes, No], Wave 3: [No, Yes]). If the cleaning engine relied on a positional index (like traditional label encoding) for transformation, this inconsistency would result in different numerical codes for the same semantic category across waves (e.g., Yes $\rightarrow$ 1.0 in Wave 1 and 0.0 in Wave 3). To overcome this, the cleaning engine was designed to be *order invariant* wherever possible, particularly in the later steps. For multi class scenarios (Step 6), this required an explicit, wave independent lookup based on the category string itself to ensure that a category string, once transformed to a numerical value, remained consistent regardless of its appearance order in the raw data.

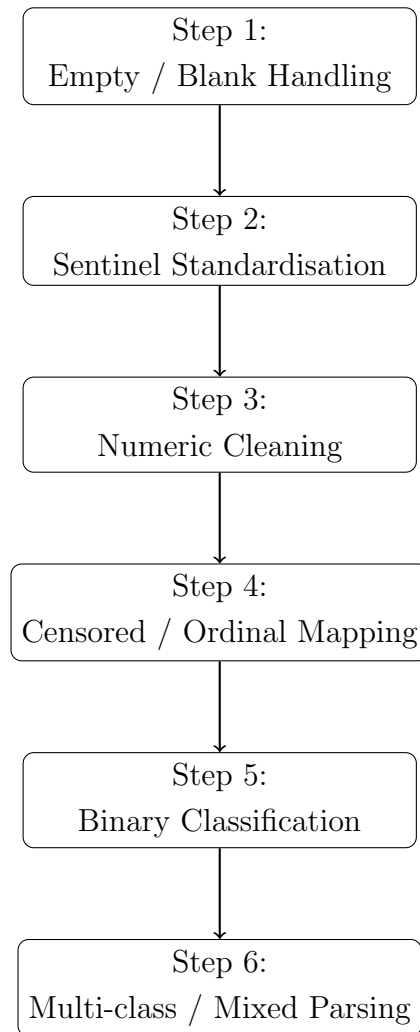


Figure 4.1: Flowchart of the Six Step Cleaning Pipeline applied to all TILDA waves. See full transformation tables in Appendix B.7 and D for detailed scenario progression code at each step.

4. IMPLEMENTATION DETAILS

The steps, and their technical focus, are:

1. **Empty / Blank Values:** Converts all NaN, empty strings, and whitespace to the standard non response value `-1.0`.
2. **Sentinel Values:** Standardises wave specific non response codes (e.g., `-98`, `-99`) and documented textual sentinels (e.g., `"dk"`, `"not applicable"`) from the TILDA PMF release guide to the unified `-1.0` placeholder.
3. **Numeric Transformation:** Removes non numeric characters (units, symbols, white spaces) and converts censored or annotated numeric strings (e.g., `"50+"`, `"<10"`, `"1-2.5"`) to a numerical float equivalent (e.g., boundary value or midpoint).
4. **Censored Categorical:** Applies explicit, known ordinal transformations and pre defined mappings to variables where text labels represent a latent numerical or ordered scale (e.g., health rating scales, likert ranges). This step was intentionally performed manually for a subset of low count variables to build intuition and reliable mapping dictionaries for the more complex subsequent scenarios. This included conversions for calendar based frequency strings like `"twice a week"` to a numerical weekly equivalent.
5. **Binary Categorical:** Automatically maps two level variables (e.g., Yes/No, Male/Female) to `1.0/0.0`. Importantly, this mapping is *order invariant*: the numerical assignment does not depend on the original ordering of category labels in the raw data (e.g., `[Male, Female]` versus `[Female, Male]`). Instead, the encoding relies on *semantic detection rules* (e.g., assigning `0.0` to negated or absence indicating strings such as `"no"`, `"not"`, or wave specific refusal codes, under context constraints) alongside *count based detection rules* (where a variable exhibits exactly two non missing unique values). This design choice ensures consistent binary representation across waves and variables, independent of label ordering or survey specific value presentation.
6. **Multi-class / Mixed (The Complex Scenario):** This final step processes all remaining multi label, mixed type, and string numeric hybrid

4.2 Data integration and longitudinal alignment

columns. The challenge was more complex by the fact that the order of categorical string values for the same variable could differ across waves in some cases. To handle this, numerical assignments were made using an explicit dictionary lookup based on the original order sample value category strings, rather than relying on positional or alphabetical order. This ensured consistent numerical representation. The step performs extensive lexical processing, dedicated number extraction, and rule based mapping to convert dense, unstandardised text into numerical floats, ensuring all residual variable types are successfully quantified. The transformed values were subject to increased manual verification due to the variable complexity.

4.2 Data integration and longitudinal alignment

Following cleaning, features were aligned across waves based on:

Chapter 5

Empirical Evaluation

5.1 References

Open the file ‘references_myphd’ with BibTex. The file is located in ‘/7_references/’

The file contains a series of templates for the proper formatting of references according to the Harvard referencing style. Click on each reference in the text below to see how Latex has formatted the reference on the References section.

- **Book:** (? , p. 10) (page number required only if referring to an in-text quote)
- **Book Chapter:** (?)
- **Book Contribution:** (?)
- **MUSIC CD/DVD:** (?)
- **Journal Article:** (?)
- **Patent:** (?)
- **PhD/Master Dissertation:** (?)
- **Conference Paper:** (?)
- **Technical Report:** (?)

- **Unpublished Report:** (?)
- **Web Videos**(?) (. . . not the name of the person who uploaded the video though)
- **Web Paper:** (?)
- **Web Journal Article:** (?)
- **Webpage:** (?)
- **Downloaded Images or Sounds:** (?)

You can also use this kind of referencing if needed in the text:
Surname [?] said that . . .

Chapter 6

Conclusion

6.1 Summary

6.2 Conclusions

6.2.1 Data availability Statement

The data analysed in this study is subject to the following licenses/ restrictions:
The datasets generated during and/or analysed during the current study are not publicly available due to data protection regulations but are accessible at TILDA on reasonable request. Requests to access these datasets should be directed to <https://tilda.tcd.ie/data/accessing-data/>.

6.2.2 Ethics Statement

The studies involving human participants were reviewed and approved by Faculty of Health Sciences Research Ethics Committee at Trinity College Dublin, Ireland. The patients/ participants provided their written informed consent to participate in this study.

6.3 Contributions

6.4 Future Work

Appendix A

Python code for exporting PMF metadata to CSV

```
1 import pandas as pd
2 import pyreadstat
3 from pathlib import Path
4 import json
5
6 data_files = {
7     "wave1.json": "data/0053-01_TILDA_Wave1_2009-2011/0053-01
8         _TILDA_Wave1_Data/SPSS/0053-01_TILDA_Wave1_PMF_v1.12.sav",
9     "wave2.json": "data/0053-01_TILDA_Wave2_2012-2013/Data
10         /0053-02_TILDA_Wave2_PMF_v2.7.sav",
11     "wave3.json": "data/0053-01_TILDA_Wave3_2014-2015/Data
12         /0053-04_TILDA_Wave3_PMF_v3.6.sav",
13     "wave4.json": "data/0053-01_TILDA_Wave4_2016/Data/0053-05
14         _TILDA_Wave4_PMF_v4.4.sav",
15     "wave5.json": "data/0053-01_TILDA_Wave5_2018/Data/0053-06
16         _TILDA_Wave5_PMF_v5.5.sav",
17     "harmonized.json": "data/0053-06_TILDA_Harmonized_2016/Data
18         /0053-03_TILDA_Harmonized_v1.2.dta",
19 }
20
21 def load_df_with_var_of_interest(
22     var_dir="var_of_interest", data_files=data_files, verbose=
23     False
```

```

18 ):
19     dfs = {}
20     var_dir = Path(var_dir)
21
22     for json_name, data_path in data_files.items():
23         file = var_dir / json_name
24         with open(file, "r") as f:
25             vars_list = json.load(f).get("variables_of_interest")
26             if verbose:
27                 print(f"Variables_of_interest_from_{file.name}:_{vars_list[:10]}_...")
28
29             if data_path.endswith(".sav"):
30                 df_full, meta = pyreadstat.read_sav(
31                     data_path, encoding="latin1", apply_value_formats=True
32                 )
33                 df, _ = pyreadstat.read_sav(
34                     data_path,
35                     usecols=vars_list,
36                     encoding="latin1",
37                     apply_value_formats=True,
38                 )
39             else:
40                 df_full = pd.read_stata(data_path,
41                                         convert_categoricals=True)
42                 df = pd.read_stata(data_path, convert_categoricals=True)[vars_list]
43
44             if verbose:
45                 print(
46                     f>Data_shape_(Original):_{df_full.shape[0]}_rows_x_{df_full.shape[1]}_columns"
47                 )
48                 print(f>Data_shape_(Cleaned):_{df.shape[0]}_rows_x_{df.shape[1]}_columns")
49                 print("
50                     -----")
51
52     dfs[file.stem] = df

```

Appendix

```
52     return dfs
53
54
55 if __name__ == "__main__":
56     for json_name, path_str in data_files.items():
57         path = Path(path_str)
58         wave_name = json_name.replace(".json", "")
59         if path_str.endswith(".sav"):
60             # SPSS
61             df, meta = pyreadstat.read_sav(
62                 path, encoding="latin1", apply_value_formats=True
63             )
64
65         else: # Stata .dta file
66             df, meta = pyreadstat.read_dta(path, encoding="latin1")
67
68         pd.DataFrame(
69             {"variable": meta.column_names, "label": meta.
70              column_labels}
71         ).to_csv(Path("output") / f"{wave_name}
72                  _variable_descriptions.csv", index=False)
73
74         print(f"-----wave_{path}-----")
75         print(df.shape)
76         print(df.columns[:20])
77         print(
78             "Variable/Feature_descriptions, csv_file_saved_
79             successfully->dir=output/"
80         )
81         df.to_csv((Path("output") / f"{wave_name}.csv"), index=
82                   False)
83
84     dfs = load_df_with_var_of_interest(verbose=True)
```

Appendix B

Scenario Cleaning Pipeline Tables

Table B.1: Initial Raw Feature Counts Categorised by Nine Scenarios (Pre Cleaning)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	9	0	7	0	9	0
Numeric with sentinel	187	59	143	60	100	141
Clean numeric	14	27	18	22	20	8
Censored / bracketed numeric	84	71	91	85	85	73
Binary categorical	248	159	177	220	207	187
Multi class categorical / mixed	267	166	203	205	156	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total Columns	809	482	639	592	577	453

Appendix

Table B.2: Scenario Count Progression: Post Step 1 (Empty / Blank Handling)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	81	31	42	32	75	15
Clean numeric	124	55	123	50	54	134
Censored / bracketed numeric	84	71	91	85	85	73
Binary categorical	251	159	179	220	207	187
Multi class categorical / mixed	269	166	204	205	156	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	809	482	639	592	577	453

Table B.3: Scenario Count Progression: Post Step 2 (Sentinel Standardisation)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	205	86	165	82	129	149
Censored / bracketed numeric	84	71	91	85	85	73
Binary categorical	251	159	179	220	207	187
Multi class categorical / mixed	269	166	204	205	156	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	809	482	639	592	577	453

Table B.4: Scenario Count Progression: Post Step 3 (Numeric Cleaning)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	263	117	215	124	172	149
Censored / bracketed numeric	16	23	17	16	17	73
Binary categorical	251	160	180	221	207	187
Multi class categorical / mixed	279	182	227	231	181	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	809	482	639	592	577	453

Table B.5: Scenario Count Progression: Post Step 4 (Censored / Ordinal Mapping)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	279	140	233	140	189	222
Censored / bracketed numeric	0	0	0	0	0	0
Binary categorical	251	160	180	221	207	187
Multi class categorical / mixed	279	182	226	231	181	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	809	482	639	592	577	453

Table B.6: Scenario Count Progression: Post Step 5 (Binary Classification)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	530	300	413	361	396	409
Censored / bracketed numeric	0	0	0	0	0	0
Binary categorical	0	0	0	0	0	0
Multi class categorical / mixed	279	182	226	231	181	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	809	482	639	592	577	453

Table B.7: Final Dfs Cleaned: Post Step 6 (Multi-Class / Mixed Parsing)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	809	482	639	592	577	453
Censored / bracketed numeric	0	0	0	0	0	0
Binary categorical	0	0	0	0	0	0
Multi class categorical / mixed	0	0	0	0	0	0
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	809	482	639	592	577	453

Appendix C

First Stage: Scenario Variable Mapping

```
1 # Sentinel values (numeric and string)
2 # -1 and -1.0 is sentinel but will be used in the clean pipeline
3 standardised_sentinels = {'-1', '-1.0', -1, -1.0} # set ->
    duplicates removed -> counts correct -> single value detection
    is zero
4 numeric_sentinels = [-1, -1.0, -98, -98.0, -99, -99.0,
5                      95, 95.0, 96, 96.0, 98, 98.0, 99, 99.0]
6 string_sentinels = [str(x) for x in numeric_sentinels] + [f"_{x}"
7                  for x in numeric_sentinels] + \
8                  ['na', 'n/a', 'refused', 'missing', 'dk', 'rf', 'refused', "
9                  refuse", 'none', 'no_response', 'nan', 'not_answered', 'no
10                 _answer',
11                 'not_applicable', 'no_consent', 'dont_know', "don't_know", "didn
12                 _t_answer", 'prefer_not_to_say', 'task_not_done', '
13                 insufficient_sample', 'unable_to_record',
14                 '__', '*']
15 blanks = ['', ' ', '\t', '\n']
16
17 SCENARIOS = [
18     "Empty/_blank_values",
19     "Numeric_with_sentinel",
20     "Clean_numeric",
21     "Censored/_bracketed_numeric",
22     "Binary_categorical",
```

```

18     "Multi_class_categorical/_mixed",
19     "Single_value_categorical",
20     "Special_symbol/_mixed",
21     "Unknown/_complex_mixed"
22 ]
23
24
25 # flatten all dtypes across the waves to float and category
26 def flatten_column_types(df):
27     df_flat = df.copy()
28
29     for col in df_flat.columns:
30         if pd.api.types.is_numeric_dtype(df_flat[col]): # Numeric
31             columns
32             df_flat[col] = df_flat[col].astype(float) # Convert
33             to float
34         else:
35             df_flat[col] = ( # Convert to string, strip
36                 whitespace, lowercase, then category
37                 df_flat[col]
38                 .astype(str)
39                 .str.strip()
40                 .str.lower()
41                 .astype('category')
42             )
43
44     return df_flat
45
46 def detect_column_scenario_mapping(df, sample_size=10):
47     summary = []
48
49     # Regex patterns
50     special_symbols_pattern = r'^A-Za-z0-9\s\.,\-\+&/()' #
51         special symbols excluding common numeric formatting
52     bracketed_numeric_pattern = r"\d{1,3}(?:,\d{3})*[-/]\d{1,3}(?:,\d{3})*|\d+\s*[\<>+-]" # patterns like "10-20",
53         "30/40", ">=50"
54     sentinel_set = set(x.lower() for x in string_sentinels) #
55         Lowercase string sentinels for comparison
56
57     for col in df.columns:

```

C. FIRST STAGE: SCENARIO VARIABLE MAPPING

```
52     col_data = df[col].copy()
53     dtype = str(col_data.dtype) # get data type as string
54
55     # Convert to string for sentinel / blank detection
56     str_vals = col_data.astype(str).str.lower().str.strip() #
        lowercase and strip whitespace
57     clean_vals = col_data[~str_vals.isin(sentinel_set) & ~(
        str_vals.isin(blanks))] # remove sentinels and blanks
58
59     # Remove sentinels and blanks for scenario detection only
60     vals_no_sentinel = clean_vals[~clean_vals.astype(str).str
        .lower().isin(sentinel_set)
61                                     & ~clean_vals.astype(str).
        str.lower().isin(blanks)
        ].unique()
62     num_unique_no_sentinel = len(vals_no_sentinel)
63
64     # Check if any sentinel values are present
65     sentinel_present = str_vals.isin(sentinel_set).any()
66
67     # Detection flags
68     empty_detected = str_vals.isin(blanks).any() or len(
        col_data) == 0
69
70     # Numeric detection: coerce to numeric, NaNs for non-
        numeric
71     numeric_vals = pd.to_numeric(clean_vals, errors='coerce')
        # coerce non-numeric to NaN
72     numeric_like = numeric_vals.notna().all() and len(
        clean_vals) > 0 # all values are numeric-like after
        cleaning
73
74     sentinel_detected = str_vals[~str_vals.isin(
        standardised_sentinels)].isin(sentinel_set).any() # or
        col_data.eq(-1.0).any() # check for sentinel presence
75     censored_detected = clean_vals.astype(str).str.contains(
        bracketed_numeric_pattern).any() # check for bracketed
        numeric patterns
76
77     # Binary categorical detection (exactly 2 unique values
        after cleaning)
```

```

78     binary_categorical_detected = False
79     if dtype == 'category':
80         if num_unique_no_sentinel == 2:
81             binary_categorical_detected = True
82         elif num_unique_no_sentinel == 1 and sentinel_present
83             :
84             binary_categorical_detected = True
85
86     # Multi-class categorical (3+ unique values, object/
87     # category)
88     multi_class_categorical_detected = num_unique_no_sentinel
89     > 2 and dtype == 'category'
90
91     # Single value categorical
92     single_value_detected = num_unique_no_sentinel == 1 and
93     not binary_categorical_detected
94
95     # Special symbols detection (ignore numeric formatting)
96     symbol_detected = clean_vals.astype(str).str.contains(
97         special_symbols_pattern).any()
98
99     # Mixed types detection
100     mixed_detected = len(set(type(v) for v in col_data)) > 1
101     # more than one data type present
102
103     # Determine scenario in order of pipeline
104     if empty_detected:
105         scenario = SCENARIOS[0]
106         handling = "Replace numeric/categorical/empty/blank
107         with -1.0"
108     elif numeric_like and sentinel_detected:
109         scenario = SCENARIOS[1]
110         handling = "Replace numeric/sentinel values with -1.0
111         "
112     elif numeric_like and not sentinel_detected:
113         scenario = SCENARIOS[2]
114         handling = "Keep as numeric"
115     elif censored_detected:
116         scenario = SCENARIOS[3]
117         handling = "Convert numeric ranges to midpoints or
118         keep as categorical"

```

Appendix

```
110     elif binary_categorical_detected:
111         scenario = SCENARIOS[4]
112         handling = "Encode as 0/1 or keep as category"
113     elif multi_class_categorical_detected:
114         scenario = SCENARIOS[5]
115         handling = "One-hot encode, label encode, or keep as
116                     category"
117     elif single_value_detected:
118         scenario = SCENARIOS[6]
119         handling = "Keep as category or constant column"
120     elif symbol_detected:
121         scenario = SCENARIOS[7]
122         handling = "Inspect and preprocess based on context"
123     else:
124         scenario = SCENARIOS[8]
125         handling = "Inspect manually or complex mixed types"
126
127     sample_values = list(col_data.unique()[:sample_size]) #
128                     sample unique values
129
130     summary.append({
131         'Variable': col,
132         'DataType': dtype,
133         'UniqueValues': num_unique_no_sentinel,
134         'Scenario': scenario,
135         'SuggestedHandling': handling,
136         'SampleValues': sample_values
137     })
138
139     return pd.DataFrame(summary)
```

Appendix D

Second Stage: Deterministic six step cleaning engine

```
1 # Cleaning pipeline to all waves
2 dfs_cleaned = {}
3
4 # flattened dtype df
5 dfs_flat = {wave: flatten_column_types(df) for wave, df in dfs.
6             items()} # flatten dtypes
7
8 # Original df
9 display(scenario_count_table(dfs_flat).style.set_caption("
10     Original Table - pre cleaning")) # display original scenario
11     count table
12
13 # Step 1: Clean empty/missing values
14 dfs_step1 = {wave: clean_empty_missing(df) for wave, df in
15             dfs_flat.items()} # apply step 1 cleaning
16 display(scenario_count_table(dfs_step1).style.hide(axis="index").
17         set_caption("Step 1: Empty / Blank post cleaning"))
18
19 # Step 2: Clean sentinel values
20 dfs_step2 = {wave: clean_sentinel(df) for wave, df in dfs_step1.
21             items()} # apply step 2 cleaning
22 display(scenario_count_table(dfs_step2).style.hide(axis="index").
23         set_caption("Step 2: Sentinel post cleaning"))
24
```

D. SECOND STAGE: DETERMINISTIC SIX STEP CLEANING ENGINE

```
18 # Step 3: Numerical Transformation
19 dfs_step3 = {wave: numerical_transform(df) for wave, df in
    dfs_step2.items()} # apply step 3 cleaning
20 display(scenario_count_table(dfs_step3).style.hide(axis="index").
    set_caption("Step3: Numeric Transformation-post cleaning"))
21
22 # Step 4: Categorical Transformation for censored scenario
23 dfs_step4 = {wave: categorical_transform_S4(df) for wave, df in
    dfs_step3.items()} # apply step 4 cleaning
24 display(scenario_count_table(dfs_step4).style.hide(axis="index").
    set_caption("Step4: Category Transformation_S4-post
    cleaning"))
25
26 # Step 5: Categorical Transformation for binary values
27 dfs_step5 = {wave: categorical_transform_S5(df) for wave, df in
    dfs_step4.items()} # apply step 5 cleaning
28 display(scenario_count_table(dfs_step5).style.hide(axis="index").
    set_caption("Step5: Category Transformation_S5-post
    cleaning"))
29
30 # Step 6: Categorical Transformation for multi class/mixed values
31 dfs_step6 = {wave: categorical_transform_S6(df) for wave, df in
    dfs_step5.items()} # apply step 6 cleaning
32 display(scenario_count_table(dfs_step6).style.hide(axis="index").
    set_caption("Step6: Category Transformation_S6-post
    cleaning"))
33
34 dfs_cleaned = dfs_step6
35
36
37 def show_grouped_vars_for_scenarios(dfs, scenarios_to_show): #
    Function to show grouped variables per scenario across waves
    -> markdown format
38     """
39     Used to inspect the scenario variables/features that need
    handling for transformation/cleaning.
40     A pre and post cleaning helper function to identify variables
    that need specific handling and to
41     Inspect actual changes for a scenario from previous cleaning
    step to current step.
42     """
```

```

43
44 print("#_Category_Transformation_Mapping")
45 print(f"##_Scenario:_{scenarios_to_show}") # Markdown mapping
    dictionary header
46 for wave, df in dfs.items():
47     print(f"\n###_=====_{wave.upper()}_
        =====")
48
49     mapping_df = detect_column_scenario_mapping(df) # Get
        scenario mapping
50
51     for scenario in scenarios_to_show:
52         vars_in_scenario = mapping_df[mapping_df['Scenario']
            == scenario]['Variable'].tolist() # Variables in
            scenario
53         # print(f"\n--- Scenario: {scenario} ---") # Scenario
            header for pipeline git commit
54         print(f"\n---_Total:_{len(vars_in_scenario)}_---") #
            \n
55         if not vars_in_scenario:
56             print("No_variables_found.")
57             continue
58
59         # Dictionary: frozen set(values) -> list of variables
60         groups = {}
61
62         for var in vars_in_scenario:
63             unique_vals = list(dict.fromkeys(df[var].dropna()
                )) # Preserve first-seen order
64             key = tuple(unique_vals) # Use ordered tuple as
                key
65
66             if key not in groups: # Initialise list if key
                not present
67                 groups[key] = []
68             groups[key].append(var) # Append variable to the
                group
69
70         # Print grouped variables in markdown format friendly
            for Notion -> mapping dictionary
71         for value_set, var_list in groups.items():

```

D. SECOND STAGE: DETERMINISTIC SIX STEP CLEANING ENGINE

```
72         print("\n*Variables:*","" + ",".join(var_list)
73               + "'') # \n
74     print(f"*Pre_transformed_values:*_{list(value_set
75           )}')")
76
77     # Pull transformed values from dfs_cleaned /
78     specific step using preserved order
79     try:
80         df_before = df[var_list[0]]
81         df_after = dfs_cleaned[wave][var_list[0]]
82
83         # Build transformed list in same order as
84         original values
85         transformed_vals = []
86         seen = set()
87
88         for orig_val in value_set: # Iterate through
89             original values in order
90             mask = df_before == orig_val
91             if mask.any():
92                 trans_vals = df_after[mask].
93                     unique().tolist() # Get unique
94                     transformed values
95                 if trans_vals:
96                     transformed_vals.append(
97                         trans_vals[0]) # Append
98                     first transformed value
99
100     except Exception:
101         transformed_vals = []
102     print(f"*Transformed_numeric:*_{'{transformed_vals
103           }'}")
104
105     scenarios_to_show = SCENARIOS
106     # scenarios_to_show = ["Clean numeric"]
107     # scenarios_to_show = ["Censored / bracketed numeric"]
108     # scenarios_to_show = ["Binary categorical"]
109     # scenarios_to_show = ["Multi class categorical / mixed"]
110
111     show_grouped_vars_for_scenarios(dfs_cleaned, scenarios_to_show) #
112     prev step == comparison to current step else current step
```

```
transformed/cleaned for ready .md dictionary
```

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